

Mindfulness-Based Interventions

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Mindfulness refers to the non-judgmental acceptance and awareness of present experiences. Kabat-Zinn defined mindfulness as “paying attention in a particular way: On purpose, in the present moment, and nonjudgmentally” (1994, p. 4)”. Bishop et al. (2004) identify two key components of mindfulness: (1) the improved self-regulation of attention, which involves both sustaining and switching attention, and (2) an openness for and acceptance of any current experience. Mindfulness is typically trained during meditation sessions, where people are encouraged to attend to present experiences, such as their breathing, emotions, or sounds in the environment. These experiences should be merely observed, acknowledged, and accepted as they arise, without passing judgment. While mindfulness and mindfulness meditation have a long history in Buddhist tradition, it was not until the last quarter of the 20th century that they stirred interest in Western societies for the treatment of mental disorders. Since the development of Kabat Zinn’s (1982, 1990) Mindfulness Based Stress Reduction (MBSR) program, mindfulness has claimed a prominent spot in the treatment and study of different psychological problems and disorders. In the present chapter, we will first briefly outline the most common mindfulness-based therapeutic interventions for mental disorders. Next, we will review the clinical outcomes of mindfulness-based interventions (MBIs). Finally, we will discuss the limitations of the current state of the field and point to key directions for future research.

Popular Mindfulness-Based Interventions

Mindfulness-Based Stress Reduction

Kabat-Zinn’s (1982, 1990) MBSR program is arguably the most well-known mindfulness intervention. The program is typically taught in an eight to ten week course involving weekly group sessions, audio-guided homework assignments, and one full-day mindfulness retreat. During the group sessions, participants learn and practice different meditation techniques, do yoga exercises, and have group discussions. The most common meditations are the sitting

meditation, in which attention is focussed on the breathing, and the bodyscan, in which attention is sequentially focused on different parts of the body. When the mind wanders during meditation, participants are asked to acknowledge the distraction without judgment and then to redirect their attention to the focus of the meditation (e.g., the breathing). Originally developed as an intervention to aid people suffering from physical problems or chronic pain, MBSR has been used to treat a wide variety of physical and psychological problems.

Mindfulness-Based Cognitive Therapy

Based on the MBSR program, Mindfulness-Based Cognitive Therapy (MBCT) combines aspects of mindfulness meditation with principles from cognitive (behaviour) therapy. The MBCT program typically consists of eight weekly group sessions, a one-day retreat, and homework assignments. It was originally developed to prevent relapse of depression (Segal, Williams, & Teasdale, 2002, 2013). In addition to the skills also taught in MBSR, MBCT for depression focuses on automatically activated negative thoughts, emotions, and cognitions that are thought to be at the root of relapse. Patients learn to recognize these thoughts, emotions, and cognitions, and are encouraged not to react on them but to observe, experience, and accept them without judgment. More recently, researchers and clinicians have started to adapt the MBCT protocol in order to treat or prevent relapse of bipolar disorder (e.g. Williams et al., 2008), generalized anxiety disorder (GAD: e.g. Evans et al., 2008), social anxiety disorder (e.g. Van Bockstaele & Bögels, 2014), panic disorder (e.g. Kim et al., 2010), post-traumatic stress disorder (PTSD: e.g. King et al., 2013), obsessive compulsive disorder (OCD: e.g. Hertenstein et al., 2012), substance use (e.g. Bowen et al., 2009), and externalizing problems, autism spectrum disorder (ASD), and attention-deficit/hyperactivity disorder (ADHD: e.g. Bögels, Hoogstad, van Dun, de Schutter, & Restifo, 2008; van der Oord, Bögels, & Peijnenburg, 2012; de Bruin, Blom, Smit, van Steensel, & Bögels, 2015).

Application of Mindfulness in Mental Health Care

With the growing popularity of mindfulness in mental health care, many clinics offer MBSR or MBCT programs. These programs are typically offered in groups of up to 10 patients, in 8 weekly sessions of 2 to 3 hours. Groups can be heterogeneous, composed of patients with different mental disorders (for example internalising disorders), or more homogeneous, consisting of patients with one or one group of mental disorders (for example depression). Groups may also be organised around roles, such as mindful parenting groups, consisting of parents of children with mental disorders and/or parents with mental disorders. In youth mental health care, groups of children may be paralleled by a mindful parenting group for their parents (e.g. Bögels et al., 2008). MBIs are also offered individually to patients, couples, or families. The group or individual mindfulness training is guided by a trained mindfulness facilitator who has experience with the patient group. Next to the typical 8-week mindfulness training, mindfulness centres may offer short retreats or booster sessions in which patients who have followed the 8-week intervention come together to continue and deepen their mindfulness practice.

MBIs typically consist of four elements: (I) psycho-education about how attentional processes are related to psychopathology and how mindfulness can change these attentional processes and thereby change the course of psychopathology, (II) meditation and other experiential practices to experience how attentional processes operate and to train the self-regulation of attention, (III) inquiry after meditations to increase awareness of experiences and to learn about habitual patterns that maintain psychopathology, and (IV) homework registration and practice that is to be discussed in the following session. In the next section, we will give an example of each of these four elements of a mindfulness training, written from an experiential perspective.

Psychoeducation Example

“Let us start by thinking about a simple question: Do you know what you had for breakfast this morning? Even if you still remember what you had for breakfast this morning, did you really taste your food, were you really there while eating? Most of the time we find ourselves paying attention to other things than our breakfast, such as a conversation, the newspaper, our mobile phone, thoughts about the upcoming workday or about bringing the children to school, etc. Most of the time, we are what we call “multitasking”: Our attention is focussed on other things, in the past or the future, rather than on our breakfast. We are “on automatic pilot”, as eating our breakfast is a daily recurring activity that does not require much present moment attention. While being able to do things on automatic pilot is a helpful and adaptive mechanism, it may also be at the heart of our suffering and our psychopathology. While not paying attention to eating our breakfast, our minds may be caught in thinking, and much of our thinking will eventually end up in thinking about problems. Because we are doing so in an automatic way, we are not deliberately thinking about those problems and trying to resolve them. As a result, the problem will not be solved and we may feel bad. Also, while not really paying attention to our breakfast, we will not taste our food nor enjoy it (if its good). Caught in our thinking, we may not really pay attention to how we are or to the ones around us if we have breakfast in company. If we were to mindfully sit at the breakfast table, seeing, smelling, feeling, tasting, and eating our food with full attention, how would that change the course of events? How would this affect our mood, our relationship with ourselves and those around us? How would this affect what and how much we would eat? Could changing such a simple habitual pattern affect the problem for which you decided to participate in a mindfulness course, whether it is depression, stress, anxiety, sleeping, eating, ADHD, autism?”

Meditation Practice Example

“This practice heightens your attention for present moment experiences by focusing on the breath. Sit comfortably on a chair, with your back straight and free from the back of the chair. Your back, neck, and head are aligned. Upright but not stiff. Your feet are flat on the floor. Relaxed shoulders, arms at your sides, hands on your legs, an open chest. A position that embodies dignity, openness, and alertness. You may close your eyes if that feels comfortable, or let your gaze rest in one place for you. I invite you to focus your attention on your breathing. Try to become aware of your breathing moving in your belly, at other places in your body, your chest, your nose. Notice what you feel in your body with each inhalation and each exhalation. You do not need to feel anything special. Just the inhalation and the exhalation. There is no need to control your breath in any way. Let the breath breathe itself. Allow yourself to notice what you feel in your body right now, during every breath that you take. Occasionally you will find that your attention is distracted from the breathing: To thoughts, plans, reminders, or daydreams. This is what the mind does. No need to blame yourself. Simply note where the mind has wandered to, and then kindly direct your attention back to the breath. The mind may constantly wander. So every time your attention wanders, bring your attention back to the breath. ... [silence] ... Continue to focus your attention on what you feel in your body when breathing. Whenever you notice that your attention has wandered, give yourself a compliment because you have become aware of you mind wandering, take a moment to notice where it has wandered to, and then bring your attention back to the breath.”

Inquiry Example

“What did you experience during this sitting meditation? You may have experienced some restlessness. Where did you feel such restlessness? You may have felt it in your body: Restless legs, faster breathing, a tendency to get up. You may have felt it in your mind: A busy mind, full of thinking about the next moments of the day, and maybe even thoughts about how

useless the meditation exercise is and how you should spend your time on more important things. And if you would not be sitting here in this mindfulness class, you would have gotten up to do the things that are on your mind. What, if anything, have you learned by remaining seated, watching the business of mind and body? How did the restlessness develop from moment to moment? Were there more intense and less intense moments of restlessness? Were there moments of calmness between the episodes of restlessness? How was it like to feel restless? Is this a familiar feeling for you, was it different now compared to other times? Could you, sometimes, embrace your restlessness rather than fight it? Is there anything you want to explore further in your next sitting meditation, anything you want to try out in your daily life with respect to restlessness?”

Homework Example

“The homework for the coming week, until we see each other again, is to mindfully eat the first bite of every breakfast. When you sit with your breakfast in front of you, check in a moment (how am I, how is my body, how is my mindstate), and when you are ready start eating your breakfast as if this is the first time ever you had breakfast. Look with full attention at your food, colours, shapes, smell the food, feel it, notice what is going on in your body, and then slowly taste your first bite. Note each morning in your diary how this experience felt, so that we can discuss it during next week’s session. The second homework practice is to do the sitting meditation once every day. Choose a moment of the day that you are most likely to actually do it and build it in your routine of the day. Find a place to sit quietly and that supports your practice. You can use the audio recording of this meditation. Write a note in your diary each time you have practiced the sitting meditation, reminding you how it felt, so that we can share experiences next week. Remember you do not need to like it, you just need to do it.”

Effects of Mindfulness-based Interventions on Psychological Disorders

In this section of our chapter, we will address the clinical impact of MBIs on psychopathology. We will broadly follow the structure of the DSM-5 (American Psychiatric Association, 2015) to cover the entire spectrum of psychopathology. When available, we focus on the results of meta-analyses. If no meta-analyses are available, we focus on carefully designed Randomized Controlled Trials (RCTs) to illustrate results of more emerging fields. We do not discuss pilot studies (with $n \leq 15$ per group), case studies, or preliminary findings. Although such studies or findings are often innovative and may give important indications, they should be replicated in high-quality RCTs before MBIs in a certain domain can be considered as promising.

Neurodevelopmental Disorders

The chapter in the DSM-5 on neurodevelopmental disorders covers a wide variety of psychopathological phenomena, including intellectual disabilities, communication and language disorders, ASD, ADHD, learning disorders, motor disorders, and tic disorders. Although some attempts have been made to use mindfulness interventions in individuals with intellectual disabilities, learning disorders, and tics, these studies are either underpowered case studies or pilot studies lacking control groups (for a review, see e.g. Hwang & Kearney, 2013).

In the field of ASD, Spek, van Ham, and Nyklíček (2013) assigned 41 adults diagnosed with autistic disorder, Asperger syndrome, or pervasive developmental disorder to either a MBCT-group or a waiting list control group. After nine weeks of mindfulness training, patients in the MBCT-group showed decreased levels of anxiety, depression, and rumination, and increased levels of positive affect in comparison with the control group. Promising as these results are, it should be stressed that this study used a no-treatment control group rather than an active or placebo control group, which makes it difficult to assign the treatment effects to MBCT. Also, although the study suggests that mindfulness can help to reduce comorbid anxiety and

depression, there is at present no evidence in support of the idea that MBIs have an impact on the core symptomatology of ASD.

Given the crucial role of problematic attentional control in the symptomatology of people with ADHD, mindfulness training intuitively seems an appealing intervention in this population. There are several recent studies assessing the effects of MBIs in ADHD-samples (for a review, see Mitchell, Zylowska, & Kollins, 2015). These studies typically show positive effects of MBIs on ADHD symptomatology, but it should be noted that the large majority of them are small-sized pilot studies or lack control groups. In this regard, the study of Schoenberg et al. (2014) is an exception. They compared 23 adult participants diagnosed with ADHD who followed a MBCT program with a similar sample of 19 adults in a waiting list control group. The results showed reduced hyperactivity and inattention scores in the MBCT-group while the control group showed no changes. In another study, a mindfulness intervention adapted to the executive functioning problems of children with ADHD ($n = 22$) and their parents reduced children's ADHD core symptoms compared to a quasi-random within-subject waitlist period that included treatment as usual (van der Oord, Bögels, & Peijnenburg, 2012). These results should nonetheless be interpreted with caution as they may be driven by a placebo effect. In sum, although the results of small or uncontrolled mindfulness-intervention studies in ADHD generally show promise, at present, there are no large studies involving active or placebo control groups. Recently registered large-scale RCTs comparing mindfulness to medication (Meppelink, de Bruin, & Bögels, 2014) and assessing the added value of MBCT to treatment as usual (TAU; Janssen et al., 2015) are likely to fill this gap in the literature.

Schizophrenia Spectrum and Other Psychotic Disorders

The main focus of MBIs in schizophrenia and psychosis is on the openness and acceptance of experiences as they arise, and less so on the self-regulation of attention. Psychotic

experiences or hallucinations often cause distress, experiential avoidance, negative self-judgments, and rumination. In mindfulness training, patients learn to become more aware of such psychotic experiences and to accept them as transient experiences that do not define the self, without avoidance or rumination, hence reducing the distress caused by avoidance and rumination (Chadwick, Taylor, & Abba, 2005). Mindfulness is unlikely to reduce the core symptomatology of schizophrenia and psychosis (e.g., delusions or hallucinations), but, at best, it may help to reduce the distress and negative affect that is evoked by these symptoms.

A meta-analysis by Khoury, Lecomte, Gaudiano, and Paquin (2013) assessed the effects of a variety of interventions involving some aspects of mindfulness in schizophrenic patients. They found overall medium effects on within group comparisons of different symptom measures. However, when comparing these effects with control groups, the mindfulness-related interventions only yielded a better outcome when compared to a TAU group, but not when compared to waiting list or active treatment controls. Furthermore, the studies in this analysis varied with regard to how central the role of mindfulness was in the treatment. For instance, several of the included studies used Acceptance and Commitment Therapy (ACT; see Hayes, Luoma, Bond, Masuda, & Lillis, 2006). As such, it remains unknown to what extent the findings should be attributed to mindfulness training. Another limitation of this meta-analysis concerns the quality of the studies included in the analysis. Many of these studies were small-scale pilot studies or did not include control groups. Analyses based on the results of less reliable studies are also likely to be less reliable.

Research on MBIs in the field of schizophrenia and psychosis thus seems to be at a turning point. Several pilot studies included in the analysis of Khoury et al. (2013) have yielded promising findings. A recent small-sized trial ($n = 44$) has yielded even more encouraging findings, with patients following a combination of mindfulness, CBT, and pharmacotherapy

reporting higher quality of life and less severe negative symptoms of schizophrenia than a control group following a combination of CBT and pharmacotherapy without mindfulness (López-Navarro et al., 2015). These findings should be replicated in carefully designed RCTs.

Bipolar and Related Disorders

Bipolar disorder is characterized by alternating periods of mania or elevated mood and periods of depressed mood. During manic episodes, people with bipolar disorder experience overall high levels of arousal, often expressed as euphoria, high risk taking, high irritability, or even violence. In this paragraph, we will focus on the effects of MBIs on the manic episodes; the effects of mindfulness on depression will be discussed in the next paragraph.

The only RCT on the effects of mindfulness on bipolar disorder was conducted by Perich, Manicavasagar, Mitchell, Ball, and Hadzi-Pavlovic (2013). They assigned 95 patients fulfilling the DSM criteria for bipolar disorder to either MBCT or TAU. Over the treatment and 12-month follow-up periods, they found no group differences on the time to the first recurrence of a manic episode, on the total number of manic episodes, or on clinician-rated symptoms of mania. Patients in the MBCT group did report lower levels of anxiety than patients in the TAU group. Although these results are certainly disappointing, two limitations of this study should be noted. First, their trial was marked by substantial drop-out (28 participants did not receive the allocated intervention), reducing the statistical power of the study. Second, there were no group differences in the change in self-reported trait mindfulness over time, suggesting either that the specific MBCT-protocol did not adequately train mindfulness or that bipolar patients are not receptive for mindfulness training. These limitations notwithstanding, there is at present no evidence to suggest that mindfulness training can reduce the impact or occurrence of manic episodes, although the results of a recently registered large-scale RCT (Lahera et al., 2014) may lead to different conclusions in the future.

Depressive Disorders

As MBCT was designed for depression relapse prevention, there now is a large body of literature on the effects of MBIs on depression. MBCT was developed in light of cognitive models of depression, according to which depressive relapse results from a close connection between low/depressed mood states and negative, depression-related cognitions (Segal et al., 2002, 2013). After recovery from depression, these negative cognitions become largely inactive but they can be readily reactivated in response to relatively small changes in mood. Reactivation of negative cognitions in turn leads to negative thinking about the self, rumination, feelings of hopelessness and helplessness, and may thus escalate in a new depressive episode. Through MBCT, formerly depressed participants are trained to recognize and disengage from these dysfunctional negative cognitions and their impact.

Several meta-analyses have been published to assess the efficacy of MBIs in depression and depression relapse prevention. Hofmann, Sawyer, Witt, and Oh (2010) found that MBIs had medium sized effects on depressive symptoms in a sample of studies from mixed populations, and large effects in a subsample of studies with depressed samples. However, the analysis of Hofmann et al. does not provide effect sizes for the crucial pre-post comparison of mindfulness-versus control groups. In addition, many of the included studies were of relatively poor methodological quality, suffering from problematic randomization and/or blinding and often lacking control groups. As mentioned before, meta-analyses based on the results of methodologically shaky studies are likely to yield less reliable effect size estimates.

In a more recent meta-analysis of only RCTs ($k = 4$), Strauss, Cavanagh, Oliver, and Pettman (2014) found that MBIs had medium to large effects on depression symptoms, compared to a combination of active and passive control groups. However, this effect was in no small part driven by an extremely large effect in a study of Strauss, Hayward, and Chadwick (2012), in

which depressed patients followed a person-based cognitive therapy pilot program and not a standardized MBSR or MBCT protocol.

In terms of relapse prevention, Chiesa and Serretti (2011; for highly similar results, see Piet & Hougaard, 2011) compared the impact of MBCT in addition to TAU with TAU control groups. The addition of MBCT led to a stronger decrease in residual depressive symptoms and to a reduced relapse risk of 30 to 36% (inconsistency in paper, text p. 448 vs. Figure 2a). However, both results are based on a very small number of studies, with $k = 2$ for the residual depressive symptoms and $k = 4$ for the relapse rate. In a slightly larger sample including more recent studies ($k = 7$), Clarke, Mayo-Wilson, Kenny, and Pilling (2015) estimated that MBCT reduces the risk of relapse with 21%.

In sum, the literature shows that MBIs have a moderate effect on depression symptoms, with little or no evidence to support the idea that MBIs are more effective than other interventions. MBCT reduces the risk of relapse of depression by about 20 to 30%, which is similar to other relapse prevention programs (see Clarke et al., 2015). In a recent review, van der Velden et al. (2015) further showed that decreased depressive symptoms and relapse risks were associated with or mediated by increased trait mindfulness, suggesting that changes in mindfulness underlie the changes in depression. Of note, studies on mediating mechanisms rarely correct for other variables that may change next to mindfulness (e.g., optimism) and rarely test the exact opposite mediation, that decreased depressive symptoms mediate increased mindfulness. Taken together however, MBIs can be considered effective interventions for depression and depression relapse prevention.

Anxiety Disorders

The large majority of research on MBIs in anxiety disorders has focused on GAD and social anxiety disorder. In the meta-analysis of Hofmann et al. (2010), MBIs resulted in medium-

sized decreases in anxiety in a mixed sample of studies. In a subsample of studies ($k = 7$) with samples suffering from clinical anxiety, MBIs even had a large effect on anxiety symptoms. Although these results are positive, it should be reemphasized that this analysis did not provide pre-post comparison of mindfulness versus control groups, nor did it employ strict inclusion criteria.

In line with these limitations, the more recent meta-analysis of only RCTs ($k = 8$) by Strauss et al. (2014) yielded more modest estimates of effect sizes. They found that MBIs had a medium-sized effect on anxiety symptoms in anxious populations. However, these results should also be interpreted with caution. First, the beneficial effect of MBIs seems to be driven in part by one study with an extremely large effect. Second, the control groups in this meta-analysis were diverse, including both active and passive control conditions. In this regard, it is important to note that the two studies favouring the control group over the mindfulness group (and hence reducing the clinical impact of MBIs) used active cognitive-behaviour therapy control groups. In a combined analysis of depressed and anxious samples, Strauss et al. showed that MBIs result in larger reductions on primary symptom measures than inactive control groups, while no such differences were apparent when comparing MBIs with active control groups.

Thus, although it seems unlikely that MBIs are more effective than CBT in reducing anxiety, there is strong evidence for the idea that MBIs have a clinically significant impact on anxiety. One final limitation to this conclusion concerns the scope of mindfulness research in anxiety disorders. Most if not all RCTs have focused on GAD, social anxiety disorder, or panic disorder (for an exception on health anxiety, see McManus, Surawy, Muse, Vazquez-Montes, & Williams, 2012). We found no studies on the effects of mindfulness training on separation anxiety disorder, agoraphobia, or specific phobias or fears, although mindfulness has been theorized to increase the effects of extinction learning and exposure therapy (Teanor, 2011).

Obsessive-Compulsive and Related Disorders

OCD is characterized by recurrent and persistent thoughts, impulses, or images (obsessions) and repetitive behaviours or mental acts in response to these obsessions (compulsions). Hale, Strauss, and Taylor (2013) proposed that MBIs could prove useful in the treatment of OCD because learning to non-judgmentally accept any current experience should reduce the impact and frequency of behaviours to control obsessive thoughts. Also, by learning that thoughts are not necessarily facts, the prominence and persistence of obsessive thoughts may decrease. Finally, mindfulness training may increase patients' awareness of compulsive behaviours as well as alternative behavioural options. Although this line of reasoning is intuitively appealing, there is at present hardly any strong empirical research substantiating these ideas. Research on MBIs in OCD is limited to a few case-studies and one underpowered trial in students. The results of two recently registered RCTs (Külz et al., 2014; Strauss et al., 2015) are likely to shed more light on the potential of MBIs for OCD.

Trauma- and Stressor-Related Disorders

Although the DSM category of trauma- and stressor-related disorders also includes other disorders, mindfulness research has focused solely on PTSD. PTSD is characterized by a state of hyperarousal, flashbacks of the traumatic event, and experiential avoidance of trauma-related situations or memories. With its focus on the openness for and acceptance of any current experience, mindfulness has been hypothesized to be particularly useful in countering avoidance and reducing the impact of hyperarousal (e.g., Vujanovic, Niles, Pietrefesa, & Schmertz, 2011). In addition, the improved self-regulation of attention may help PTSD patients to flexibly switch between attending to feelings of distress and shifting attention away from ruminative thoughts (Vujanovic et al.).

Over the past decade, many pilot studies on the effects of MBIs in PTSD have been published, most of them with small samples or other methodological shortcomings (Banks, Newman, & Saleem, 2015). In a small RCT with veterans, Kearney, McDermott, Malte, and Martinez (2013) compared the effects of MBSR as an addition to TAU (n = 21) with a TAU control group (n = 23). Although they found some post-intervention and follow-up between-group differences in depression and quality of life, the authors failed to report the significance of crucial interactions, making it hard to conclude that adding MBSR improved the outcome of TAU. The only convincing study of MBIs for PTSD that we found is the RCT by Polusny et al. (2015). They assigned 116 veterans suffering from PTSD to either a standard MBSR-course or a present-centred group intervention involving psychoeducation, discussions of daily problems and difficulties, and peer support. In addition to improved mindfulness and quality of life, patients in the MBSR group showed a larger decrease in both self-reported and clinician-rated PTSD symptom severity than patients in the control group. As such, there is evidence in support of MBIs for the treatment of PTSD, although the effect sizes were modest and replication seems warranted.

Feeding and Eating Disorders

Feeding and eating disorders are associated with overeating or undereating for extended periods of time thereby overriding internal feedback signals of physical hunger and satiety (Kristeller, 2015). Furthermore, eating problems are also associated with over-reactivity to external food cues (Kristeller, Wolever, & Sheets, 2014). MBIs may address these problems as they focus on greater awareness to hunger and fullness cues, sensory-specific satiety, and emotional and other triggers for eating (Kristeller et al.). Consistent with this idea, a meta-analysis based on 24 studies concluded that attentive eating influences food intake (Robinson et al., 2013). In addition, through increased awareness and acceptance of any current sensations,

mindfulness may also reduce reactivity to feelings of craving (Witkiewitz, Lustyk, & Bowen, 2013; see also the section on substance-related disorders below).

We found two small-sized RCTs in individuals with subclinical levels of binge-eating and obesity symptoms (Daubenmier et al., 2011; Timmerman & Brown, 2012) and one medium sized RCT in a clinical population of overweight binge-eating individuals (Kristeller et al., 2014). Daubenmier et al. randomized women (M Body Mass Index (BMI) = 31.1) who wanted to control the effects of stress on eating behaviour to a 10-session mindfulness program designed specifically for stress eating ($n = 24$) or a waitlist control group ($n = 23$). While they found some effects on secondary outcome measures, they found no significant effect on body weight. Timmerman and Brown randomized women who eat out often (M BMI = 31.8) to either six 2-hour mindful eating classes ($n = 19$) or to a wait list control condition ($n = 16$). The mindfulness group subsequently showed a relatively small but significant weight loss (-1.7kg), whereas the control group showed no weight change (-0.2kg). However, at baseline, the control group (M weight = 76.4kg) in this study weighed considerably less than the mindfulness group (M weight = 91.5kg), complicating the interpretation of the findings.

The only clinical trial (Kristeller et al., 2014) focussed on binge eating. Overweight and obese individuals (M BMI = 40.3) were randomized to a 12-session Mindfulness-Based Eating Awareness Training (MB-EAT) ($n = 39$), a Psycho-Educational/Cognitive-Behavioural intervention (PECB; $n = 27$), or a waitlist control group ($n = 26$). One and four months after the intervention, both treatment groups showed large decreases in binge eating and self-report measures of craving and eating behaviour, as well as decreased feelings of depression and increased self-esteem, while participants in the waiting list control group remained relatively stable over the entire period. Differences between the two treatment groups were very limited. In a subset of individuals who met DSM-IV-R criteria for binge eating disorder at baseline, 80% of

the MB-EAT group and 82% of the PECB group no longer met the criteria for binge-eating disorder one month after the intervention, while this number was only 38% in the control group. In addition, 25% of the MB-EAT, while all participants in the control group reported bingeing episodes. Despite these encouraging findings, neither treatment had an effect on BMI, though within the MB-EAT group, more mindfulness practice was related to greater weight loss.

Overall, the effects of MBIs on feeding and eating disorders are mixed: Although the only clinical trial resulted in significant effects on self-reported binge-eating and eating behaviours, mindfulness training had no effects on BMI. Research on mindfulness in feeding and eating disorders is also limited to binge eating disorder and obesity; no research to date has addressed the impact of MBIs on other common eating disorders such as anorexia nervosa or bulimia.

Sleep-Wake Disorders

Sleep disorders can be categorized into sleep-wake disorders (insomnia, hypersomnolence, and narcolepsy), breathing related sleep disorders, and parasomnias. Effectiveness of MBIs has only been investigated for sleep onset and sleep maintenance problems, and in particular insomnia. Ong et al. (2014) compared MBSR (n = 19), mindfulness-based therapy for insomnia including specific behavioural strategies for insomnia (MBTI, n = 19), and a self-monitoring control group (SM, n = 16) in adults. Results on subjective and objective measures of sleep showed that MBSR and MBTI were equally effective and superior to SM at post-test. At 6-months follow-up, MBTI showed greater reductions in subjective insomnia scores than the MBSR group, although this difference was not replicated in the objective sleep parameters. Black et al. (2015) compared mindfulness with an extensive sleep hygiene education and relaxation programme of 6 weeks in a sample of older adults with moderate sleep disturbances (n = 49). Relative to sleep hygiene education, mindfulness had larger effects on subjective sleep quality as well as on secondary measures such as depression and insomnia.

Finally, in Chinese adults aged 75 years and older, MBSR (n = 30) significantly improved subjective sleep quality compared to a wait-list control group (n = 30) (Zhang et al., 2015). From these RCTs, it can be concluded that MBIs are an effective approach for insomnia. However, the results of subjective measures were more convincing than those of objective sleep measures and sample sizes were generally quite small.

Substance-Related and Addictive Disorders

Addiction is characterized by the failure to control desire (Widiger & Smith, 1994), as demonstrated by high relapse rates (McHugh, Hearon, & Otto, 2010). Given that mindfulness meditation was developed to counteract the adverse influence of desire on suffering (Rahula, 1959), there is intuitive appeal in using MBIs to treat addictive behaviours. Although theoretical pathways through which mindfulness may help to overcome substance-related craving have been proposed (e.g., Breslin, Zack, & McMains, 2002; Brewer, Elwafi, & Davis, 2013; Marlatt & Ostafin, 2006), there is relatively little high-quality research regarding both the efficacy of MBIs for addiction and the pathways through which mindfulness may work.

A recent systematic review found 21 studies using MBIs to treat addictive behaviours (Chiesa & Serretti, 2014). Of these, only seven were RCTs with interventions focusing primarily on mindfulness, and only two of these RCTs had medium/large sample sizes and reported follow-up assessments beyond end-of-treatment. A first study (n = 163) found that compared to TAU, a mindfulness-based relapse prevention (MBRP) programme led to lower alcohol and drug use at 2 months post-treatment, but comparable relapse at 4 months (Bowen et al., 2009). The second study, on nicotine dependence, found that compared to a standard smoking cessation programme (n = 41), participants who followed a mindfulness relapse prevention programme (n = 46) reported greater abstinence up to 3 months post-treatment (Brewer et al., 2011).

Since Chiesa and Serretti's (2014) review, the evidence base regarding mindfulness and addiction has grown with several RCTs. One study that is notable for its large sample size and length of follow-up compared MBRP (n = 91-79, depending on measurement phase) with cognitive-behavioural relapse prevention (CBRP, n = 72-64) and TAU (n = 70-64) (Bowen et al., 2014). These authors found no group differences in drug use or heavy drinking at the 3-month follow-up, but both intervention groups demonstrated less relapse at 6-months and the MBRP group demonstrated less drug use (but not fewer heavy drinking days) at the 12-month follow-up. A second study compared MBRP with CBRP (Witkiewitz et al., 2014). At a 15-weeks post-treatment assessment, the MBRP group showed fewer days of use on an aggregate of alcohol and drug use than CBRP. However, given that only 1/28 MBRP and 5/26 CBRP participants reported using either alcohol or drugs at follow-up, these results may be unreliable. Nevertheless, these small relapse numbers also indicate that both programs do seem effective in preventing relapse.

Two studies also examined MBIs for smoking. The intent-to-treat analyses of Davis et al. (2014) showed no differences on smoking behaviour between a mindfulness group (n = 105) and participants encouraged to use a self-help phone service (n = 91) at a 6-month follow-up. Among participants who actually initiated the treatments, the mindfulness group did smoke less at follow-up, although the study was marked by large attrition and baseline group differences. A second study in a small sample of individuals with mild intellectual disability found that compared to TAU (n = 19), mindfulness (n = 21) led to less smoking and increased abstinence at a 1-year follow-up (Singh et al., 2014). In sum, theory and initial results indicate that MBIs holds promise for treating addictive behaviours.

Neurocognitive Disorders

In the DSM-5, the chapter on neurocognitive disorders covers a wide variety of disorders characterized primarily by progressive deficiencies in important cognitive functions such as

attention, inhibition, (working) memory, language, and perception. In most neurocognitive disorders, the underlying pathology and aetiology is well-known (e.g., HIV/AIDS, Alzheimer's disease, Traumatic brain injury, Parkinson's disease).

In patients with HIV/AIDS, Gonzalez-Garcia et al. (2014) assigned participants to either MBCT focussing on quality of life (n = 19) or to a no-treatment control group (n = 20). Following the intervention, the MBCT group reported higher overall quality of life, less stress, and less feelings of depression and anxiety. Similar results were also obtained in a large trial by SeyedAlinaghi et al. (2012). They compared the effects of a standard MBSR course (n = 85) with a brief education and support group intervention (n = 86). After the intervention, the MBSR group reported less psychological symptoms than participants in the control group. This decrease was maintained both at 3- and 6-month follow-up, but was absent at a 9-month follow-up and even reversed at a 12-month follow-up. Finally, and rather surprisingly, Gayner et al. (2012) found that a standard MBSR group (n = 78) and a no-treatment control group (n = 39) showed similar reductions in negative affect, depression, and anxiety, although only the MBSR-group reported increased positive affect. In sum, although mindfulness does seem to have an impact on general psychological wellbeing (stress, anxiety, depression), no study to date has addressed the effects of mindfulness on the core cognitive deficits of neurocognitive disorder due to HIV infection (i.e., executive functions, motor skills, speeded information processing, episodic memory, attention/working memory, language, and perception; for a review, see Woods, Moore, Weber, & Grant, 2009).

Although it has been argued that mindfulness may help in the treatment or prevention of Alzheimer's disease (Larouche, Hudon, & Goulet, 2015), only one recent trial on this matter has been published. Quintana-Hernández et al. (2016) assigned Alzheimer's patients to either a standard biomedical treatment group (donepezil; n = 29), a mindfulness and donepezil group (n =

41), a muscle relaxation training and donepezil group (n = 38), or a cognitive stimulation and donepezil group (n = 33). Although cognitive functioning was better preserved in the mindfulness group than in the standard control group, it should be noted that these results only held for what seems to be a post-hoc and ill-defined subset of patients with mild symptoms of Alzheimer's disease. As such, we feel that there is at present no strong evidence for the effects of mindfulness on neurocognitive functioning in Alzheimer's disease. One trial on patients with traumatic brain injury found reduced feelings of depression in an MBCT-group (n = 38) compared to a waiting list control group (n = 38), although the researchers did not assess cognitive functions (Bédard et al., 2013). To our knowledge, the effects of MBIs on cognitive and psychological symptoms of Parkinson's disease have not been systematically assessed (but for a recently registered trial, see Advocat et al., 2013).

Discussion

Over the past decades, research on has grown exponentially. Reviewing the evidence from meta-analyses and RCTs (with minimal n per group > 15) for the effectiveness of MBIs in psychological disorders, we showed that there is relatively strong evidence for the effectiveness of MBIs in depression and depression relapse, anxiety disorders (mostly GAD and social anxiety disorder), sleep disorders (only insomnia), and addictions (mostly relapse prevention). There is also limited evidence for the clinical effectiveness of mindfulness in the treatment of ADHD and PTSD, but these results need to be replicated. Although MBIs can improve overall quality of life and reduce comorbid anxiety and mood problems in ASD, bipolar disorder, eating disorders (only studied in obesity), and neurocognitive disorders, there is little evidence to date to suggest that mindfulness has an impact on the *core* symptomatology of these disorders. Finally, we found no RCTs on the effects of mindfulness on OCD or other DSM-5 disorders. However, there is also

little if any theoretical basis for MBIs in these disorders (except for OCD), so the lack of RCTs in these domains should not be surprising.

Although the overall pattern of clinical effectiveness is promising, mindfulness research suffers from relatively consistent and recurring limitations. In many disorders, clinical studies are limited to RCTs with small samples and/or no treatment-control (only wait-list control) groups. Such small-scale studies can yield interesting findings and raise provocative questions for future research. However, in order to develop a clearer picture of the benefit of MBIs in clinical practice, compared to existing treatments, it will be important for future research to strive for greater methodological quality. First, in order to improve the overall reliability of the results, larger sample sizes are needed (Ioannidis, 2005), especially given the fact that most RCTs that we discussed were marked by large drop-out rates. Second, randomizing participants should be done very carefully in order to prevent the typical result of stronger findings in non-randomized trials (Colditz, Miller, & Moesteller, 1989). Third, there is a clear need to include active control groups to account for placebo effects (Baskin, Tierney, Minami, & Wampold, 2003). Fourth, follow-up measurements should be extended to longer periods (> 1 year) to provide important information about whether any benefits of MBIs are long-lasting, and whether there are “sleeping effects”, that is, effects that appear only in the long term.

Another issue that remains to be addressed in future research concerns the mechanisms driving the clinical effects of MBIs. In line with Bishop et al.’s (2004) definition of mindfulness, MBIs typically target both the regulation of attention and the non-judgmental acceptance of any current experience. However, many researchers have adapted standard mindfulness protocols in order to match the specific needs of certain populations. While it may be clinically relevant to tailor interventions, these changes also complicate the comparison of studies. Hence, while our review shows that MBIs can be effective, there is very little research on which aspects of MBIs

(improved attention regulation, increased awareness and acceptance of current experiences, or other additional factors) are actually driving these clinical changes. In the field of depression, Van der Velden et al. (2015) showed that changes in attention, memory specificity, self-discrepancy, emotional reactivity, and momentary positive and negative affect underlie the clinical effects of MBCT. In research on other disorders however, the focus of studies is often on the clinical effects and the underlying mechanisms driving the effect remain largely hypothetical and are seldom systematically addressed.

The questions whether or not, how often, and how long participants need to practice (meditate) in order to maximize the benefits (so called dose-response research) is another issue of debate and avenue for future research. Several studies have addressed this issue by correlating patients' amount of homework/meditation practice with clinical outcome measures and found a positive correlation (e.g., see Kristeller et al., 2014). However, this correlation does not necessarily prove the point, as it may well be that only those patients who experience benefits from practice continue to meditate. Williams et al. (2014) investigated this issue in a robust way by comparing psychoeducation about the principles of mindfulness in a group (with no meditations and no meditation homework practice assignments) with the full MBCT program for patients with recurrent depression. Highly experienced mindfulness trainers led both the MBCT and psycho-education groups. Aside from a superior effect of MBCT for patients with a history of childhood trauma, the psychoeducation proved equally effective. This study puts into question whether it is really needed for patients, at least those with recurrent depression, to meditate.

MBIs are now also increasingly used in the area of developmental psychopathology, that is, children with psychopathology and their parents. As most mental disorders have their onset before adulthood, it is important that mindfulness practices are adapted to fit children of different ages and developmental stages. Mindfulness in children and adolescents with anxiety disorders

and depression is in the stage of pilots, often with subclinical populations (Lee, Semple, Rosa, & Miller, 2008; Semple, Lee, Rosa, & Miller, 2010). Particularly for childhood depression, mindfulness is a promising treatment given that (1) the effect sizes of CBT for childhood depression are disappointing compared to those found in adult depression, and (2) combined with mindful parenting (e.g., Bögels et al., 2008 in youth with externalising disorders) such a combined approach could tackle family variables that have caused or maintain childhood depression (Restifo & Bögels, 2009).

Mindful parenting as a stand-alone intervention for parents of children with psychopathology (Bögels, Hellemans, van Deursen, Römer, & Meulen, 2014; Bögels & Restifo, 2014) has also been shown to effectively decrease psychopathology in children and their parents compared to waitlist controls. An RCT on the effects of mindfulness-based child birth and parenting for anxious or depressed mothers-to-be on their own psychopathology, the delivery, as well as on their babies' development is currently being conducted (Veringa, de Bruin, & Bögels, 2014). Also, mindfulness for older people suffering from psychological disorders is being piloted (e.g., Creswell et al., 2012). Thus, mindfulness is now studied in all developmental phases as well as in the family context in order to improve the parent-child interaction and thus influence the outcome in terms of developmental psychopathology. To our knowledge, these interventions have been proven feasible, some have been proven effective, while research on others is still ongoing.

Finally, next to viewing mindfulness training as a form of psychotherapy, it can be regarded as a way of living, a programme to improve life style. From that perspective, the goal of mindfulness training is not necessarily to reduce or cure psychopathology, but to improve the quality of living, by relating differently to the ups and downs of life, including psychopathology. As 38% of our population each year suffers from some sort of psychopathology (Wittchen et al.,

2011), psychopathology can be viewed as a “flu” that comes and goes, and relating in the best possible way to the symptoms can reduce suffering and increase happiness. Given the high comorbidity of mental disorders, mental disorders should be viewed as behavioral manifestations of psychopathology rather than separate diagnoses, and they should be measured dimensionally rather than categorically (e.g. LeBeau et al., 2015). Consequently, there is a need for transdiagnostic approaches to psychopathology (e.g., Harvey, Watkins, Mansell, & Shafran, 2004; Kring & Sloan, 2009), in which not one mental disorder is treated, but common underlying factors of many mental disorders are influenced. Poor attention control and inhibition are such characteristics that many disorders seem to have in common (Caspi et al., 2014), and given the focus of MBIs on both attentional control and inhibition, their potential for the treatment and prevention of psychopathology across the lifespan is large.

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